AD-YOLO: A Real-Time YOLO Network With Swin Transformer and Attention Mechanism for Airport Scene Detection

Wentao Zhou, Chengtao Cai, *Senior Member, IEEE*, Chenming Li, Hao Xu, and Haochen Shi

***Abstract*— Real-time acquisition of airport scene information is crucial for airport safety and optimization of airport utilization efficiency. However, detecting airport objects is still a challenging task due to the small size of person and vehicle targets in the airport scene images, insufficient public airport data, and so on, which makes it difficult to achieve high accuracy and real-time detection methods in the airport scene simultaneously. This article proposes a novel airport object detection approach to address the challenge by integrating the advantages of improved you only look once (YOLO), Swin Transformer, and attention mechanism [airport detector—YOLO (AD-YOLO)]. Specifically, we introduce the Swin Transformer, which retains the Transformer's ability of global attention to obtain features and reduces the drawbacks of computational complexity, into the head network based on YOLOv7 to improve the high-dimensional information feature fusion. We also design an efficient channel spatial attention (ECSA) module and introduce a small object detection layer (SODL) to improve the detection accuracy of small targets in the airport scene. We test the proposed method on the self-constructed multiple airport surveillance dataset (MASD) containing 5736 images captured by actual airport and online airport video. The experimental results show that AD-YOLO achieves 71.6% mean average precision (mAP), exceeding the mAP of the baseline method by 4.4%. The proposed method has 101.4 frames/s (FPS) on the NVIDIA RTX3080 GPU and 17.8 FPS on the Jetson Orin NX, meeting the real-time and accuracy requirements of the airport scene. Finally, the experimental results on the public airport surface surveillance (ASS) dataset show that AD-YOLO outperforms other detection methods, demonstrating its effectiveness.**

***Index Terms*— Airport scene, attention mechanism, object detection, Swin Transformer, YOLOv7.**

I. Introduction

IN INTELLIGENT air traffic control, airport monitoring object detection is an essential component of intelligent airports [1]. The airport air traffic control tower needs to obtain essential information, such as airport targets, to provide real-time status for air traffic control. It is crucial for improving airport safety, optimizing airport flight, vehicle, and personnel scheduling and achieving maximum efficiency in airport operations [2], [3], [4]. However, the number of aircraft in the airport has increased, and more aircraft have taken off and landed at airports. The airport environment is complex, aircraft operations are busy, and civil aircraft accidents, such as ground traffic conflicts and airport runway intrusions, occasionally occur, which seriously threaten aircraft safety and affect passenger travel [5], [6], [7]. Therefore, applying airport object detection technology in the intelligence air traffic control system is significant. The detection technology also has significant application value in assisting air traffic controllers and preventing aircraft conflicts, runway intrusions, and other aspects [8], [9].

Currently, research on the combination of artificial intelligence and traffic is mostly focused on vehicle detection, while little research has been done on applying object detection algorithms to airport monitoring [10]. The automatic detection method in airport monitoring is faced with the challenge of relatively small objects, such as persons and vehicles, and it is not convenient to obtain airport scene samples [11]. To address the above issues, many scholars have proposed various methods, categorized into those based on image processing and those based on deep learning, such as classification [12], segmentation [13], and so on. However, detection methods based on image processing are limited in their generalization and robustness for specific object types [14]. Each object requires a manual design of feature extraction methods and tuning of parameters. The time consumption of the detection algorithm is enormous, and it is challenging to meet the real-time demand of the industry [15]. Compared with classical feature extraction methods, the deep neural network has the characteristics of solid feature extraction ability and has achieved higher accuracy in object detection technology based on deep learning. Many scholars have shifted their research focus from traditional feature extraction methods to detection based on deep learning [16], [17], [18]. The you only look once (YOLO) algorithm has better real-time performance and accuracy than the region-convolutional neural Network (RCNN) method in many fields [19], [20], [21], [22], [23]. Therefore, this article selects the object detection method based on the YOLO framework. However, there are many small objects in the airport scene, and the small object detection ability of YOLO is weak [24]. We suggest that the Swin Transformer and attention modules with a small number

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of parameters be integrated into the network to improve the airport detection capability of the method while ensuring the real-time performance of the method.

In this article, we propose an efficient channel spatial attention (ECSA) module, which combines the efficient channel attention module (ECAM) and efficient spatial attention module (ESAM). The ECSA module structure is similar to the convolutional block attention module (CBAM) structure, which generates new features through learning channels and spatial learning of latent information, and the process is parameter-based. The input data at each level will obtain parameter fusion to obtain finer features. In addition to the contribution of the ECSA to effectively extract airport target features, we choose the Swin Transformer module, which has the feature extraction ability of a transformer module but low computational complexity, to integrate into the original network backbone. Based on the above two core components, airport detector—YOLO (AD-YOLO) can meet the requirements of airport monitoring to obtain real-time target information in real time. The main contributions of this article are summarized as follows.

1) We propose a fusion Swin Transformer and attention mechanism for the airport scene detection method, which achieves intelligent information supervision of the transportation in the airport scene. The proposed method significantly improves airport target detection performance with minimal reduction in real-time resources, which improves airport safety effectiveness, and optimizes airport utilization efficiency.
2) This article designs an ECSA module. We extend the idea of efficient channel attention for deep convolutional neural networks (ECANet) using 1D convolutional layer (Conv1D) to achieve cross-channel information exchange to spatial and design an ESAM. Then, the feature map is processed through ECAM and ESAM to obtain feature channels and spatial attention weights. Finally, the ECSA is combined into the backbone and head of the proposed method to enhance the ability to detect small objects in the airport scene.
3) There is a lack of relevant datasets for multiple airport target detection research. We use image acquisition equipment at the aircraft training base and collect multiple online airport videos. This article extracts airport images from over 200 video images and creates a multiple airport scene dataset containing 5736 images.
4) Compared with representative detection methods, the AD-YOLO method scores the highest mean average precision (mAP) in the self-constructed dataset. Then, the proposed method tests the speed of real-time detection on the Jetson Orin NX. In addition, ablation experiments are conducted to verify the effectiveness of each proposed module, and the performance of the ECSA module is separately tested at different positions. Finally, we test the generalization performance of AD-YOLO on the public airport surface surveillance (ASS) dataset.

The rest of this article is organized as follows: In Section II, the relevant work of the object detection algorithm as well as the application of the airport object detection algorithm is introduced. The proposed real-time detection framework is discussed in Section III, including the details of AD-YOLO, the Swin Transformer, and the ECSA module. Section IV focuses on the experimental setup and results analysis to demonstrate the effectiveness of the proposed method. Section V introduces the research summary and future work.

II. Related Work

A. Introduction to the Object Detection Algorithm

Object detection is a critical problem in computer vision, aiming to find the bounding boxes and object classes of objects in a given image [25]. In recent years, deep learning techniques have greatly improved object detection accuracy. The object detection method based on deep learning is typically categorized into the one-stage and two-stage detectors. The two-stage detectors propose regions and then classify objects [26]. RCNN is a typical method of two-stage detectors. It was the first attempt to utilize deep learning for image object detection. RCNN extracts features for each candidate region by the CNN, but it needs to be faster for real-time detection [27]. Fast-RCNN is faster than R-CNN because it extracts features from the entire image by the CNN and obtains features of each candidate region from the feature map [28]. Faster-RCNN improves upon Fast-RCNN by introducing the region proposal network (RPN), further enhancing speed and accuracy [29]. The one-stage detectors directly predict the categories and positions of all objects in an image. Representative methods of this approach include single shot multiBox detector (SSD), CenterNet, detection transformer (DETR), and YOLO. SSD adapts to objects of different sizes and aspect ratios by using anchor boxes at multiple scales of feature maps for detection [30]. CenterNet achieves object detection by directly predicting the center points of objects [31]. DETR captures the global relationships of objects and completes the object detection task by introducing object queries and position encodings [32]. The YOLO series detectors have undergone multiple generations of development and are considered more suitable for airport applications than other detectors.

B. Introduction to the YOLO Algorithm

YOLO is the one-stage object detection algorithm. It predicts the positions and categories of multiple bounding boxes on each grid in real time by separating images into grids [33]. Compared with YOLOv1, YOLOv2 adopts a more extensive network structure, improving the network detection performance. Because YOLOv2 can simultaneously detect 9000 different target categories, it is also called YOLO9000 [34]. YOLOv3 introduces three different detection layers responsible for detecting targets of various scales and uses the Darknet53 backbone network. These changes have significantly improved network detection performance and speed [35]. YOLOv4 designs a new backbone network named CSPDarknet53. It introduces innovative technologies such as the complete-IOU (CIOU) loss function and path aggregation network structure to optimize model performance [36]. YOLOv5 uses a simplified and efficient network structure. Although there is no article, it has a good test effect on the

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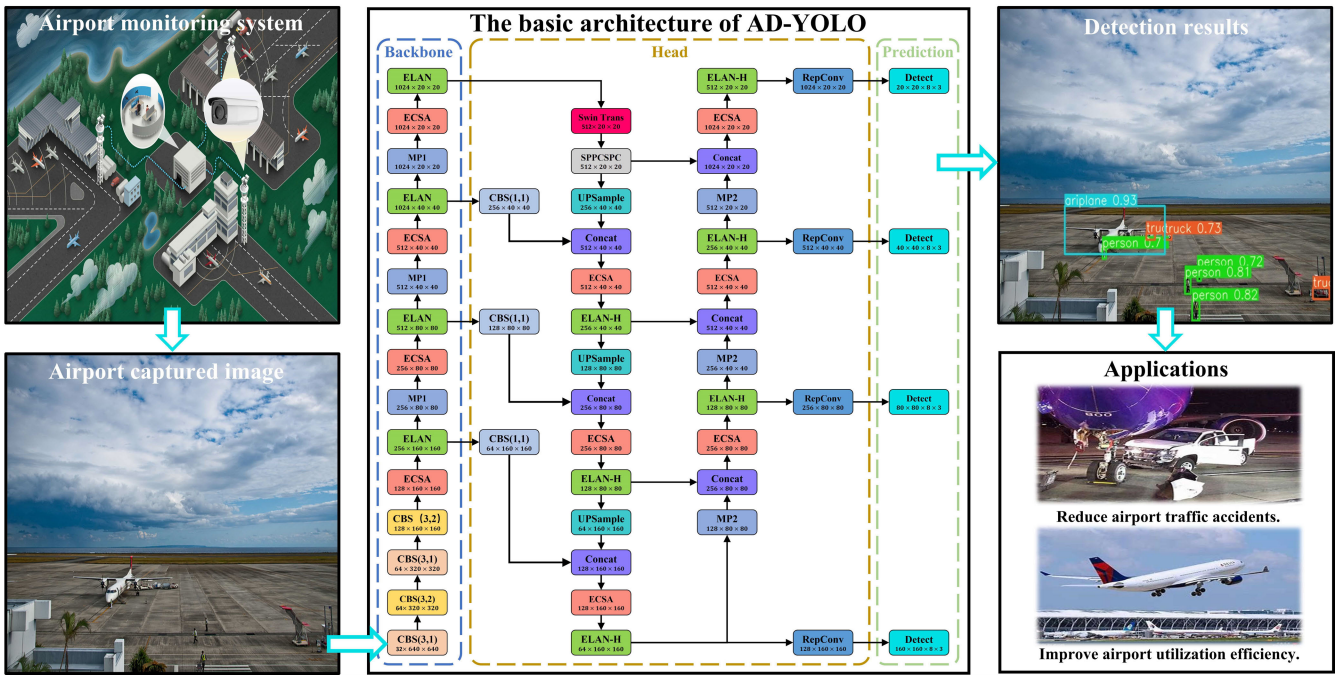


Fig. 1. Overview of the airport detection framework. First, the airport monitoring system obtains airport-captured images through airport-monitoring equipment. Then, the airport-captured image is subjected to AD-YOLO to obtain the detection results. Finally, airport management can reduce airport traffic accidents and improve airport utilization efficiency based on the detection results.

MSCOCO dataset [37]. According to the current research, YOLOv7 outperforms all known object detection methods in speed and accuracy from 5 to 160 frames/s (FPS) [38]. Therefore, this article selects YOLOv7 as the baseline method in our experiments. YOLOv8 adopts the Anchor-Free algorithm. Inspired by efficient layer attention network (ELAN) in YOLOv7, it designs a CSPDarknet53 to 2-Stage FPN (C2f) structure to improve detection speed while ensuring detection accuracy. However, when YOLO is used to detect small objects in the airport scene, the limited resolution of small objects can lead to error detection and missing detection problems.

C. Application of the Airport Object Detection Algorithm

In recent years, researchers have conducted research in the airport using object detection methods to address existing problems. To address the safety hazards of foreign object debris on airport runways, Zhang et al. [39] optimized the YOLOv5 algorithm by replacing the cross stage partial network (C3) module with the C2f module, introducing attention mechanisms and the SCYLLA-IoU loss function. The proposed algorithm achieved a 5.4% mAP improvement over the original YOLOv5. To ensure the rapid detection of airport runway damage for aircraft safety, Liang et al. [40] accelerated training by using lightweight operations in the feature extraction process and reduced model complexity by removing redundant nodes in the feature fusion process. The proposed method outperformed advanced state of the art (SOTA) models in a self-made dataset. Zhang et al. [41] applied unmanned driving technology in volunteer actions on airport grounds and proposed an unmanned ground support equipment detection and control framework. They used lightweight YOLO to detect boundary lines of channels and ensure precise management of vehicle posture and speed under various apron conditions. Wang et al. [42] proposed an airport detection network based on adaptive spatial feature fusion for airport remote sensing image detection. The adaptive spatial fusion strategy enhanced the detection capability of airport objects by improving the scale invariance of network features. The mAP of the original YOLOv5 increased by 3.6% in the self-made airport remote sensing data. Li et al. [43] researched airport detection in remote sensing the real-open world. A multicore binary analysis method was constructed based on the global development of regional, remote sensing image airport sample dataset. Although there are many studies on airport-related detection, there is limited research on the general type of airport object in airport scenes. The main reason is the lack of reliable airport scene image data.

III. Method

A. Network Architecture of AD-YOLO

According to the available research, there are many small objects in the airport surveillance image [44]. In this article, AD-YOLO is a fast and suitable object detection model for airport scenes, which can achieve advanced test results. It is based on the YOLOv7 structure and is easy to use and expand. The airport-monitoring system obtains airport scene images through cameras. Airport controllers use the AD-YOLO method to detect common airport targets and assist with their work. The overview of the airport detection framework is shown in Fig. 1. The network structure of AD-YOLO consists of the backbone and head network. The backbone network is used to extract different scale features from the input image. Then, the head network uses the feature fusion method to enhance the feature extraction. Finally, the results of classification confidence and location regression are obtained.

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In the backbone network, the downsampling and channel adjustment of the given input image are completed by different convolution, batch normalization, SiLU (CBS) modules. Specifically, the CBS module with a convolution kernel of 1 and step size of 1 retains the original shape of the picture and integrates different channel information. The CBS module with a convolution kernel of 3 and a step size of 1 does not change the size of the feature map and deepens the number of feature map channels. The CBS module with a convolution kernel of 3 and a step size of 2 can halve the shape of the feature map while extracting the feature. Then, the backbone network feature extraction is completed after the ECSA and ELAN modules. In the head network, the feature fusion method is used to enhance the relevant features of the airport object. By introducing the Swin Transformer module and the small object detection layer (SODL), the proposed method is a technology that can effectively improve the feature extraction of network small objects. Then, we design the ECSA module to enhance the focus of fusion features on airport objects and continue to improve the detection effect of the proposed method. The remainder of this section describes the network improvements and small object detection enhancement methods in detail.

B. Swin Transformer Block

In the airport scene, extracting necessary features from global features using only the CNN is difficult, while the transformer model can integrate features at different positions in the image. The transformer model has achieved remarkable success in natural language processing (NLP) due to its powerful processing capabilities and flexibility [45]. It has also been attracting the attention of researchers in computer vision to apply this model to computer vision tasks. In adapting the transformer from language to vision, the main challenges faced are the large variation in the scale of visual entities, the performance of the visual transformer may not be excellent in different scenarios, and the high resolution of the image leads to a significant increase in computational complexity. The Vision Transformer (ViT) model proposed by Google in 2020, which performs well on the image classification task, has become a milestone work in applying the transformer in computer vision [46]. The Microsoft research team proposed the Swin Transformer, a generalized backbone network in computer vision [47].

The Swin Transformer is a hierarchical model with shifted windows. Due to the hierarchical structure, the model is very flexible and can provide feature information of various scales. The Transformer module is based on global computational attention, so it has a very high computational complexity. However, the Swin Transformer limits the attention calculation to each window, improving computational efficiency and making computational complexity grow linearly for image size. The Swin Transformer blocks based on this window partitioning approach can be formulated as follows:

$$\hat{z}^{l} = \text{W-MSA}(\text{LN}(z^{l-1})) + z^{l-1} \tag{1}$$

$$z^{l} = \text{MLP}(\text{LN}(\hat{z}^{l})) + \hat{z}^{l} \tag{2}$$

$$\hat{z}^{l+1} = \text{SW-MSA}(\text{LN}(z^{l})) + z^{l} \tag{3}$$

$$z^{l+1} = \text{MLP}(\text{LN}(\hat{z}^{l+1})) + \hat{z}^{l+1} \tag{4}$$

where $\hat{z}^{l}$ and z^{l} represent the outputs of the shifted window multihead self-attention (SW-MSA) module and the multilayer perceptron (MLP) module of the block. As the patch merges at each stage, the receptive field expands after every stage, thus approximating the computation of global attention. The architecture of the Swin Transformer is shown in Fig. 2. The Swin Transformer introduces a sliding window operation to better interact with adjacent windows. The feature information of adjacent windows can be obtained by sliding windows, indirectly achieving the effect of global modeling. Currently, the versatility and flexibility of the Swin Transformer allow it to be seamlessly integrated into various vision tasks. In standard benchmark tests such as ImageNet image classification [48], MSCOCO object detection [37], and ADE20k semantic segmentation [49], the Swin Transformer has achieved the latest best results, proving its powerful feature acquisition capability. Therefore, we add the Swin Transformer to the head of YOLOv7 to enhance the network feature extraction capability.

C. Efficient Channel Spatial Attention Module

The person and vehicle targets at the airports appear as lower pixels in surveillance images. As the YOLOv7 backbone network deepens, the features of small targets may be ignored or forgotten by the network, resulting in decreased accuracy in airport scene detection tasks. The attention mechanism enhances the accuracy of small objects by assigning greater weights to a particular class of targets and reducing irrelevant information. CBAM is a module that combines channel and spatial attention mechanisms to enhance the expressive of a network [50]. ECAM is a lightweight attention mechanism that improves network performance by eliminating dimensionality reduction operations to enhance interchannel dependencies [51]. Inspired by CBAM and ECAM, the ESAM is designed and combined with the ECAM to construct the ECSA module, as shown in Fig. 3. Specific spatial information receives more attention by increasing the weights of spatial dimensions, effectively improving the network performance. The core of ECSA updates the weights of feature channels and spatial dimensions through Conv1D operations. The feature map $F \in R^{C\times H\times W}$ as input feature is fed into the ECSA. The 1-D feature map $W_c \in R^{C\times 1\times 1}$ is obtained during the ECAM stage. The 2-D feature map $W_s \in R^{1\times H\times W}$ is obtained during the ESAM

$$F_c = W_c(F) \otimes F \tag{5}$$

$$F_s = W_s(F_c) \otimes F_c \tag{6}$$

where $\otimes$ represents the multiplication, F_c represents the outputs of the ECAM, and F_s represents the outputs of the ESAM.

The ECAM helps the model focus on channel information by channel weights, as shown in Fig. 4. First, the original feature map is compressed from $[C, H, W]$ to $[C, 1, 1]$ through adaptive global average pooling operation to obtain the initial channel weights, where C represents the number of channels. H and W represent the height and width, respectively. Then,

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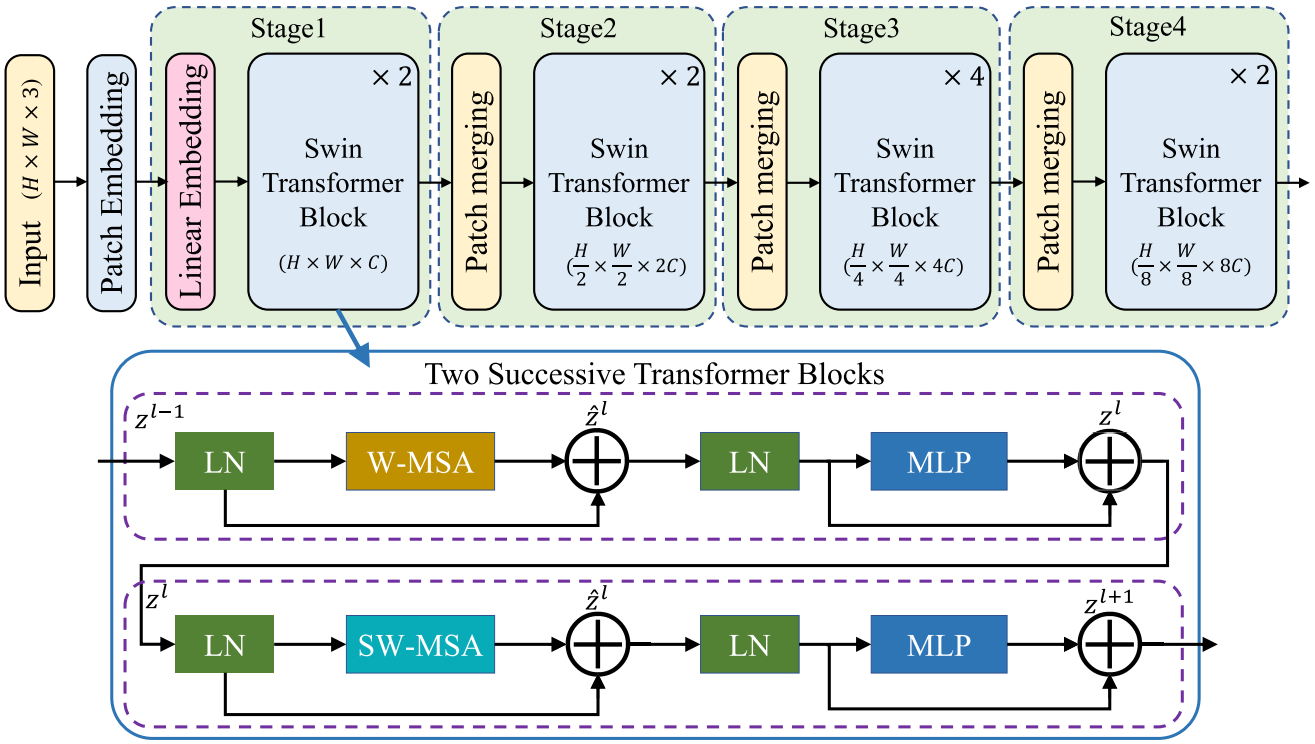


Fig. 2. Architecture of the Swin Transformer block. The input image is divided into nonoverlapping fixed-size windows. The Swin Transformer block captures local information by the window multihead self-attention (W-MSA). Subsequently, the window is shifted to achieve cross-window information exchange by using the SW-MSA. Each block contains multihead self-attention and MLP modules and gradually integrates global features through a hierarchical structure.

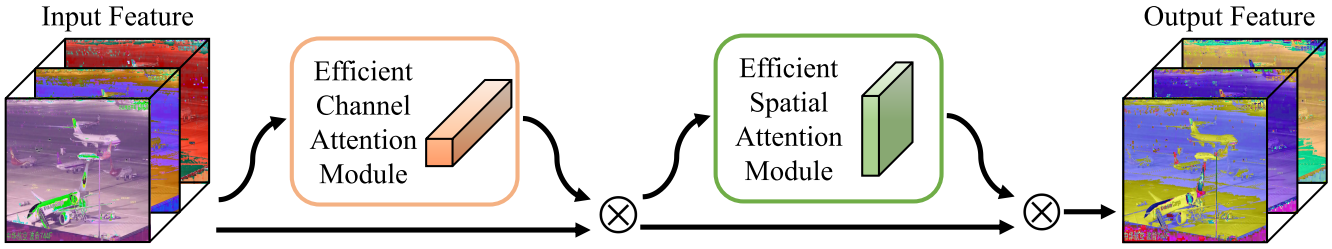


Fig. 3. Structure of the ECSA.

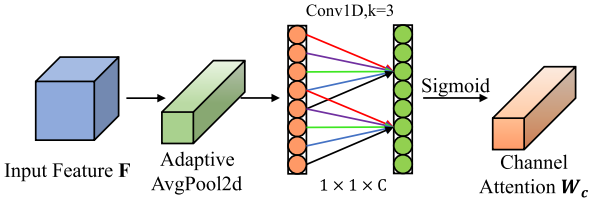


Fig. 4. Structure of the ECAM.

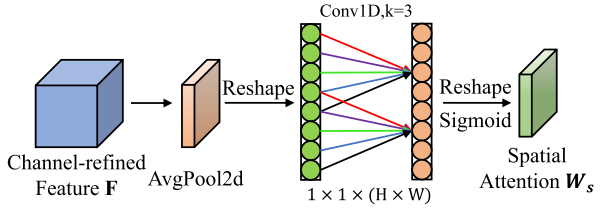


Fig. 5. Structure of the ESAM.

the channel weights are updated by Conv1D operation. Finally, the sigmoid activation function obtains the channel dimension weight of feature maps. The ESAM utilizes spatial weights to help the model focus on the regions of interest, as shown in Fig. 5. First, based on the channel attention mechanism, an average pooling operation is performed in the channel dimensions to compress the feature map from $[C, H, W]$ to $[1, H, W]$ to obtain the initial spatial weights. Then, the initial spatial weights are reshaped from $[1, H, W]$ to $[H \times W, 1, 1]$. Subsequently, the spatial weights are updated by Conv1D operation. Finally, the sigmoid activation function obtains the spatial dimension weights of feature maps. By introducing the ECSA attention module, the original YOLOv7 can better focus on the information from different channels and spatial dimensions of the feature map to improve the detection accuracy of airport small targets.

D. Small-Object Detection Layer

The original YOLOv7 provides three different detection layers of various scales for detecting small, medium, and large

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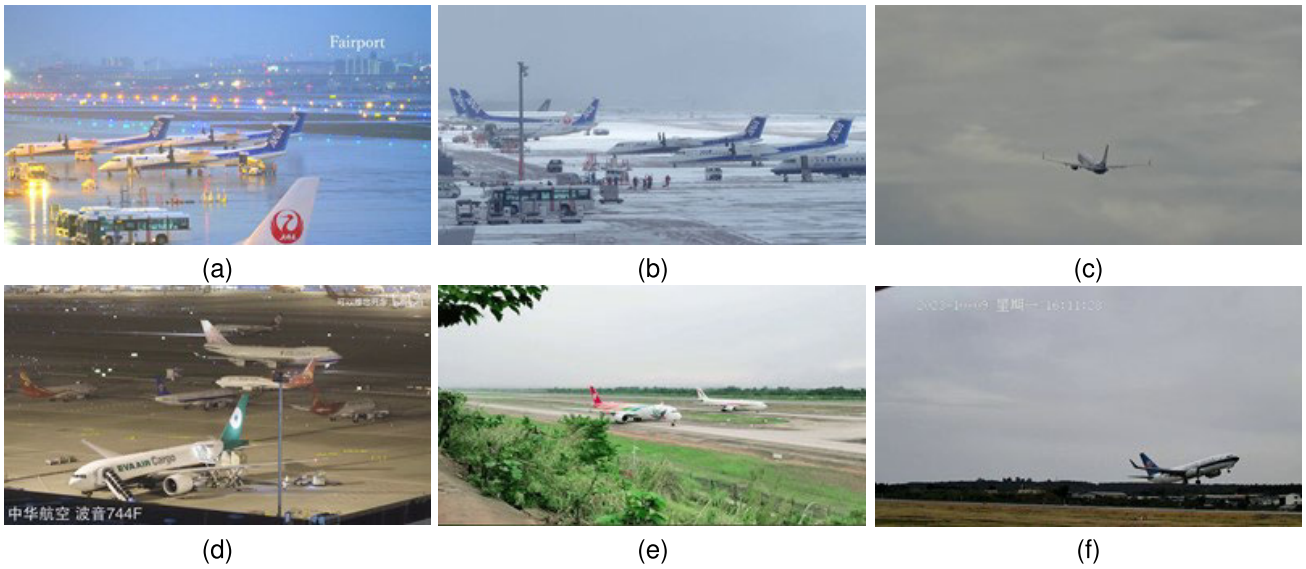


Fig. 6. Example images in the MASD. (a) Rainy airport image. (b) Snowy airport mage. (c) Foggy airport image. (d) Night airport image. (e) and (f) Captured airport image.

size of objects. The person and vehicle in the airport scene are low pixels, and after many times of downsampling, the object information in the feature map becomes very sparse. The original three detection layers can no longer satisfy small object detection in the airport scene, making it easy to miss. Therefore, we propose constructing an SODL for the network to pay more attention to small objects, which improves the detection performance [52]. The SODL is introduced into the network to specialize in detecting smaller objects in airport scene images [53]. The feature map of this layer retains more information about the location of small objects and more prosperous and more powerful detailed features, which can better obtain the location information of small objects. The specific approach is to upsample the feature map of ELAN-H and then fuse it with the first ELAN in the backbone network to form a new feature map. Finally, a new SODL is obtained after the ELAH-H module. Meanwhile, because the anchor box-based algorithm is more sensitive to the initial anchor box scale, a new anchor box scale is generated for the dataset by the K-means clustering algorithm [54]. It can effectively reduce the situation where the objects in the dataset are small and the anchor boxes are large, causing the objects to be ignored. This method reduces the missing rate and false detection rate. Introducing the SODL improves the accuracy of small object detection, making the algorithm better suited to airport scenes.

IV. Experiment

In this section, we first introduce the self-constructed multiple airport surveillance dataset (MASD). Then, we evaluate the performance of the proposed AD-YOLO in the MASD compared to SOTA methods. In addition, we conduct ablation experiments to analyze the specific contributions of each module to the AD-YOLO method. Finally, this article compares the proposed method with advanced detection methods on the public ASS dataset to verify the generality of it.

A. Multiple Airport Scene Dataset

Research on the combination of image video and artificial intelligence has been applied in various industries, while relatively little research has been done on applying detection methods to airport scene images. The main reason is the lack of relevant datasets. Sufficient training samples are the basis for model training, and a suitable dataset is crucial for airport object detection research. Although public airport datasets for the airport scene are available, the data only capture images in one or a few airport surveillance videos, which may lead to overfitting the trained network model. To address this limitation, the MASD consists of many actual and online airport images. The experiment involves image acquisition equipment at aircraft training bases to record aircraft takeoff and landing videos. We also collect online airport videos. This article collects typical airport data from more than 200 airport videos to ensure the diversity of data, including rain, snow, fog, night, and other environments. Finally, the MASD collects images every 5–20 s and manually filters and removes similar images from airport scene videos. The MASD has 5736 images, including 2548 actual images and 3188 online images. The sample images are shown in Fig. 6.

This article uses the Labeling tool to manually label the images and draw bounding boxes in three categories: aircraft, person, and vehicle. A total of 13 576 objects are labeled in MASD, with aircraft accounting for 77.5%, person accounting for 5.9%, and vehicle accounting for 16.6%. Fig. 7(a) shows the distribution of object locations, and Fig. 7(b) shows the distribution of object sizes. The distribution plot shows that small objects occupy the majority of the data, which is consistent with the characteristics of airport scene images. If the aspect ratio of the object to the original image is less than 0.1, it is considered a small object [55]. The proportion of small objects in the MASD is 61.1%. All input image shapes are set to unify training and reduce training burden. We divide the dataset into training and validation sets in the ratio of 8:2.

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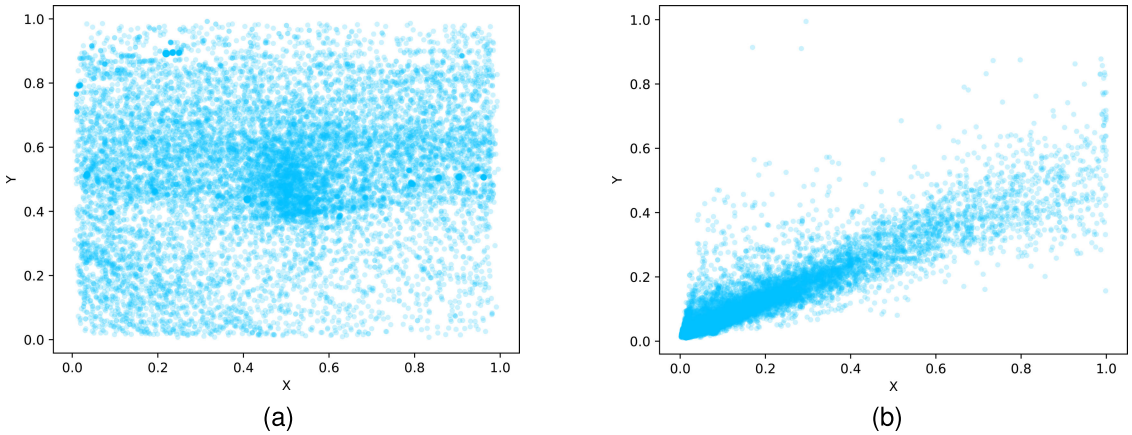


Fig. 7. Statistical visualization of MASD. (a) Distribution of objects' center points. (b) Distribution of object sizes. The darker the color of the point, the more the quantity of objects in (a) and (b).

B. Experimental Settings

1) Experimental Configurations: The Sugon cloud platform, which is equipped with Hygon C86 7185 32C CPU, 256 GB of memory, and NVIDIA RTX3080 GPU, and the Jetson Orin NX, which is powered by an Arm Cortex-A78AE v8.2, 8 GB of memory, and a 1024-core NVIDIA Ampere architecture GPU, are used in our experiment. AD-YOLO is developed based on PyTorch. During the cloud platform training process, we set the input image size to 640 × 640 and the training epoch to 300 to ensure fairness in the experiment, which may not be the optimal parameter. The Adam method (initial learning rate 0.001 and weight decay 0.0005) is the optimizer. The network is trained with a batch size of 8. Additional hyperparameters are the same as YOLOv7.

2) Evaluation Metrics: In the MASD, this article selected the precision, the recall, the average precision (AP), the mAP, and FPS as indicators to evaluate and compare the performance of AD-YOLO and other methods [56].

Accuracy and recall rates are defined as

$$\text{Racall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

where TP represents the correctly detected quantity of positive samples, FP represents the incorrectly detected quantity of positive samples, TN represents the correctly detected quantity of negative samples, and FN represents the incorrectly detected quantity of negative samples. If the IoU value of the predicted box and the actual box is greater than the threshold value, the TP quantity increases by 1.

AP is an effective indicator for evaluating the performance of the model, and AP0.5 − 0.95 is used as the main indicator in MSCOCO tasks. However, AP0.5 − 0.95 is too strict in airport detection, so the mAP in this article uses IoU = 0.5 to distinguish whether pixels are correctly predicted. AP metric is calculated as

$$\text{AP}_i = \int_0^1 P_i(R_i) dR_i \tag{9}$$

TABLE I

COMPARISON OF REPRESENTATIVE METHODS ON THE SUGON CLOUD PLATFORM

Model	Precision	Recall	aircraft	person	vehicle	mAP	FPS
Faster-RCNN	38.6	54.5	88.9	16.5	42.1	49.2	10.5
SSD300	90.9	32.1	88.0	17.5	43.7	49.6	29.1
DETR	28.7	37.4	70.7	4.3	12.3	29.1	13.7
CenterNet	72.1	37.1	90.8	24.9	67.1	60.9	17.9
YOLOv3	73.7	65.6	91.7	40.2	70.2	67.3	105.5
YOLOv4	72.6	65.8	92.1	40.3	70.9	67.7	109.9
YOLOv5l	76.4	64.2	92.0	32.4	73.1	65.9	96.0
YOLOv7	74.7	63.7	91.7	39.1	70.8	67.2	111.1
YOLOv8x	71.2	63.8	92.2	34.9	71.8	66.3	83.3
ASSD-YOLO	69.9	66.4	90.8	44.0	72.6	69.1	98.7
AD-YOLO	71.5	69.3	91.5	48.7	74.5	71.6	101.4

where P and R are the precision and the recall, respectively, when the number of all classes is detected as N, the mAP definition of the overall class is as follows:

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^{n} \text{AP}_i. \tag{10}$$

FPS indicates the algorithm processing speed. Its size is related to the network weight and the hardware configuration of the test equipment.

GFLOPs represent the number of floating-point operations during the forward propagation process of a model, which is an indicator of the computational complexity of the model.

C. Experimental Comparison

In this experiment, AD-YOLO is compared with SSD300, Faster-RCNN, DETR, CenterNet, YOLOv3, YOLOv4, YOLOv5l, YOLOv7, YOLOv8x, and airport surface surveillance detector-you only look once (ASSD-YOLO) in the MASD to demonstrate its effectiveness. This section mainly uses mAP and FPS to evaluate the model comprehensively. Due to limitations in computing resources, the comparative experimental results are focused on the MASD, summarized in Table I.

Table I shows that the proposed method achieves the best mAP value of 71.6% in airport scene object detection tasks. Compared with other YOLO methods, it has improved mAP

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TABLE II
COMPARISON OF REPRESENTATIVE METHODS ON THE JETSON ORIN NX

Method	Faster-RCNN	SSD300	DETR	CenterNet	YOLOv3	YOLOv4	YOLOv5l	YOLOv7	YOLOv8x	AD-YOLO
FPS	2.2	13.6	7.6	8.9	18.8	19.8	13.5	21.5	13.1	17.8

TABLE III
ABLATION EXPERIMENT OF EACH MODULE IN THE AD-YOLO

Model	SODL	Swin Trans	ECSA (Bockbone)	ECSA (Head)	ECSA	CBAM	Precision	Recall	aircraft	person	vehicle	mAP	GFLOPs
YOLOv7	✗	✗	✗	✗	✗	✗	74.7	63.7	91.7	39.1	70.8	67.2	103.8
YOLOv7-A	✓	✗	✗	✗	✗	✗	69.8	66.1	91.1	44.0	67.8	67.7	121.6
YOLOv7-B	✓	✓	✗	✗	✗	✗	73.7	65.3	90.9	42.9	70.7	68.2	136.6
YOLOv7-C	✓	✓	✓	✗	✗	✗	72.4	66.2	91.2	47.0	71.6	69.9	136.6
YOLOv7-D	✓	✓	✗	✓	✗	✗	69.2	68.1	91.2	46.2	74.1	70.5	136.6
AD-YOLO	✓	✓	✗	✗	✓	✗	71.5	69.3	91.5	48.7	74.5	71.6	136.6
YOLOv7-E	✓	✓	✗	✗	✗	✓	77.5	62.8	91.2	43.6	73.7	69.5	137.1

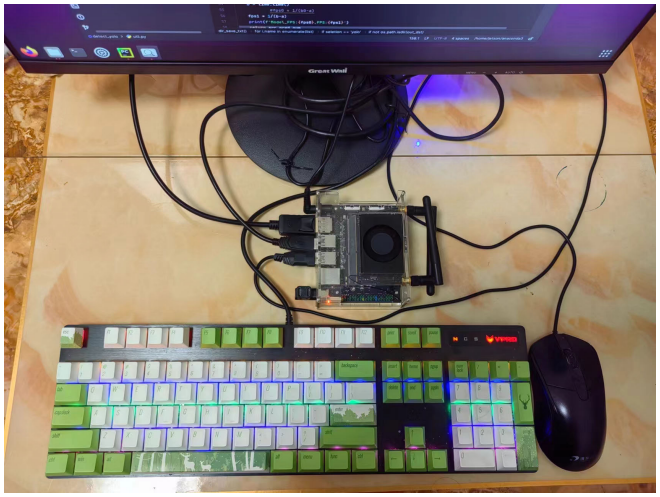


Fig. 8. Deployment platform is based on Jeston Orin NX.

by 3.9–5.7. AD-YOLO exhibits a 22% improvement in mAP over SSD, a 42.6% improvement over Faster-RCNN, a 42.5% improvement over DETR, and a 10.7% improvement over CenterNet. Compared with small object detection ASSD-YOLO, the proposed method has a 1.5% higher mAP and faster detection speed. The proposed method is most accurate in detecting small target objects such as persons and vehicles in the airport scene. The detection speed of AD-YOLO reaches 101.4 FPS, which is not the highest but meets the needs of real-time detection. Because the speed of the detection method is affected by the device hardware, we test the proposed method using the embedded Jetson Orin NX, shown in Fig. 8. We find that when the test image data and hyperparameters are consistent with the cloud platform, the embedded Jetson Orin NX evaluation metrics such as Precision, Recall, AP, and mAP are the same as the cloud platform test results, with only FPS being different. Table II shows that the AD-YOLO detection speed can reach 17.8 FPS, achieving a balance between accuracy and speed.

This article randomly selects three images and uses the above detection method. Fig. 9 shows the presence of a vehicle target in the first image. There is a problem of duplicate detection in SSD and DETR and missed detection in YOLOv3 and YOLOv4. SSD, Faster-RCNN, and DETR detect missed person targets in the second image. Although other methods have the same detection results, the proposed method has the highest overall confidence. Therefore, the AD-YOLO is more suitable for the airport scene.

D. Ablation Study

In addition, we conduct ablation experiments to demonstrate the impact of the modules introduced in the proposed method. YOLOv7-A is that adds SODL to the original YOLOv7, while YOLOv7-B is that YOLOv7 uses SODL and Swin Transformer. To explore the impact of ESCA on the method at different locations in the network, we inserted modules into the network backbone, head, backbone, and head, design YOLOv7-C, YOLOv7-D, and AD-YOLO, respectively. YOLOv7-E uses CBAM instead of ECSA in AD-YOLO. The ablation results in the MASD are summarized in Table III. We find that the GFLOPs of the model increased by 17.8 by only adding SODL. The GFLOPs of the Swin Transformer module increased by 15.0. This article continues to add that the proposed ECSA hardly increases the computational complexity of the model. The proposed ECSA module has a lightweight and efficient structure.

Table III shows that the baseline model YOLOv7 achieves 67.2% mAP. YOLOv7-A is increased by 0.5% mAP by adding SODL. YOLOv7-B is increased by 1.0% mAP by adding SODL and Swin Transformer. The Swin Transformer improves the backbone feature extraction ability, while SODL adds a structure for detecting small airport targets, effectively improving detection performance in the airport scene. We test experiments YOLOv7-C, YOLOv7-D, and AD-YOLO to verify the most suitable position of the designed attention module, which shows an improvement of 2.7%–4.4% mAP. It has been demonstrated that the proposed ECSA is better than CBAM through YOLOv7-E. The experiment shows that the attention mechanism improves the airport target detection performance of the proposed method by suppressing

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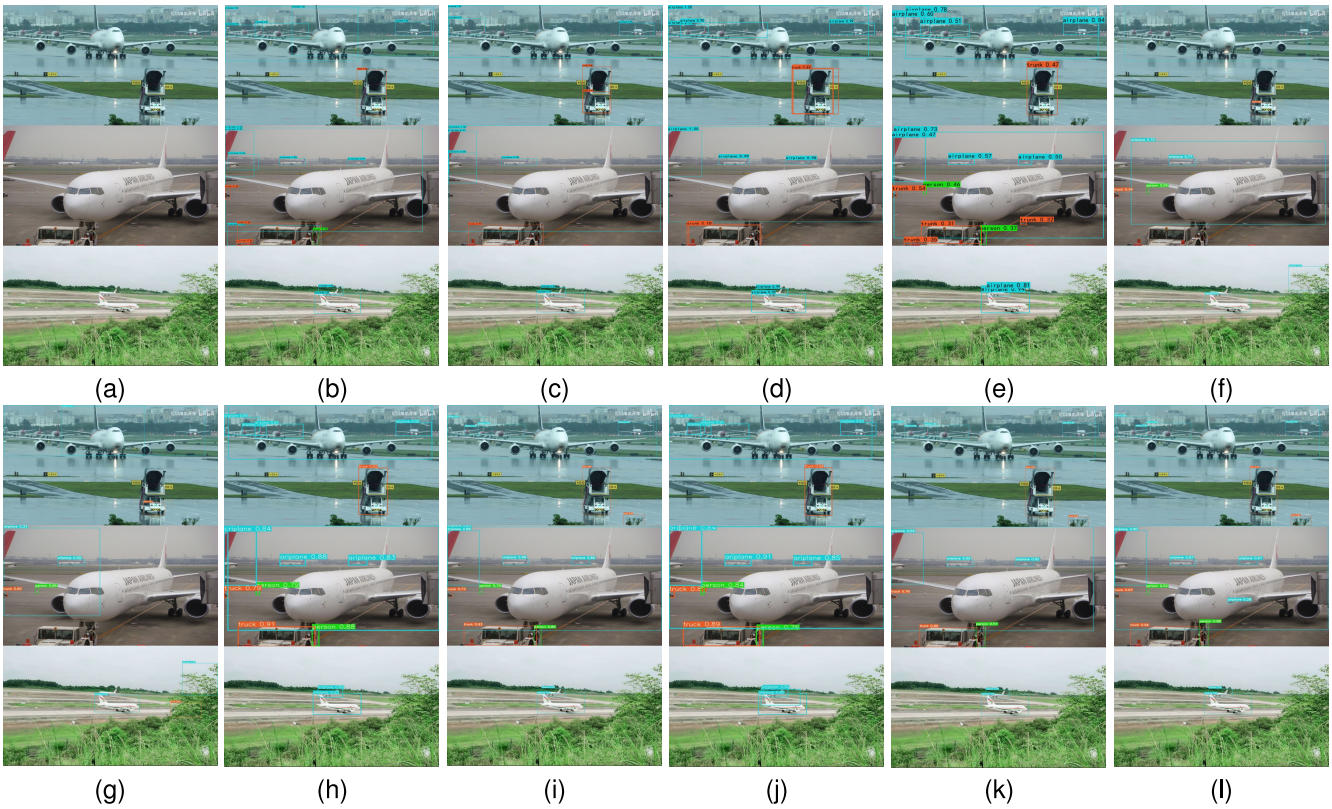


Fig. 9. Visualization of the detection results. (a) Original image. (b) Faster-RCNN. (c) SSD300. (d) DETR. (e) CenterNet. (f) YOLOv3. (g) YOLOv4. (h) YOLOv5l. (i) YOLOv7. (j) YOLOv8x. (k) ASSD-YOLO. (l) Proposed AD-YOLO.

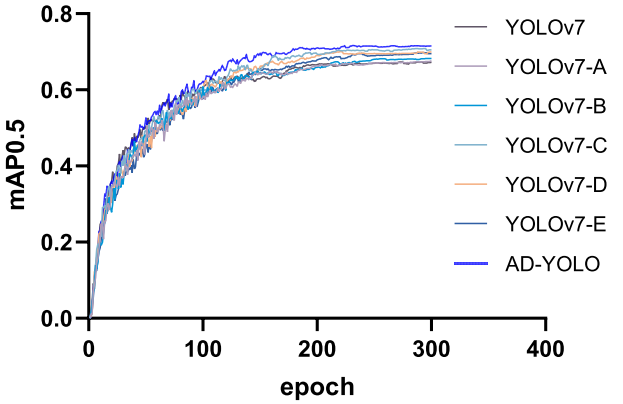


Fig. 10. Ablation experiments of each module in AD-YOLO.

small target features in background-focused images. We find that both the backbone and head of the ESCA module can improve the performance of the original model. Adding the ESCA module to both has the best effect. It indicates that the ESCA module at different positions can jointly enhance the detection performance to detect airport targets. Fig. 10 visually shows the training and testing results of each module, and each module has improved the mAP of the original YOLOv7. These results indicate the effectiveness of each module.

Image size affects the accuracy of the model in small-object detection tasks to investigate the impact of image size on the model; we set the image size to 160 × 160, 320 × 320, 480 × 480, 640 × 640, and 960 × 960 training parameters for experimentation. Table IV shows that as the image size increases, the mAP of the proposed method increases, but it also brings the problem of FPS decreasing. Therefore, the proposed method can set the corresponding image input size as needed in the application and airport detection environment.

TABLE IV
COMPARISON OF THE INPUT IMAGE WITH DIFFERENT SIZES ON AD-YOLO

Image Size	Precision	Recall	Aircraft	Person	Vehicle	mAP	FPS
160×160	92.9	18.3	62.6	0.0	6.2	22.7	357.1
320×320	66.9	53.7	88.7	20.1	62.7	57.2	294.1
480×480	67.6	64.5	91.3	37.7	70.2	66.4	208.3
640×640	71.5	69.3	91.5	48.7	74.5	71.6	101.4
960×960	78.3	66.3	90.3	50.1	75.9	72.1	68.5

E. Experimental Comparison in the Public Dataset

The public ASS dataset is used to test the universality of AD-YOLO in the airport detection task. The ASS dataset includes the airport surface surveillance dataset (ASS1) and the airport surface panoramic surveillance dataset (ASS2) [44]. The ASS1 dataset is collected by a civil aviation airport-monitoring system. It consists of 2000 images with 8371 targets, 39.4% of which are small targets. The ASS2 dataset is collected by panoramic monitoring of three civil aviation airports. It includes 100 images with 8660 targets, of which small targets account for 99.8%. We compare

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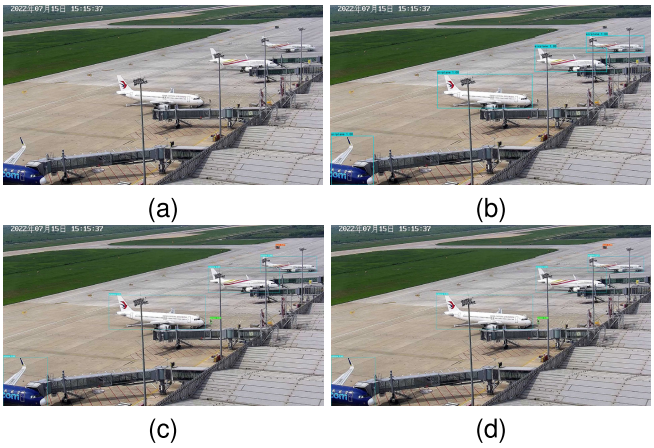


Fig. 11. Visualization of the representative detection results on the ASS1. (a) Faster-RCNN. (b) YOLOv5l. (c) YOLOv7. (d) Proposed AD-YOLO.

TABLE V
COMPARISON WITH OTHER METHODS IN THE ASS1 DATASET

Model	Precision	Recall	Aircraft	Person	Vehicle	mAP
Faster-RCNN	74.3	55.8	98.6	0.1	58.7	52.3
SSD300	85.4	32.5	90.1	7.0	46.4	47.8
CenterNet	88.2	74.2	99.4	39.0	92.3	76.9
YOLOv3	90.5	88.0	99.0	78.4	88.5	88.7
YOLOv4	89.8	84.8	99.4	74.6	90.8	88.3
YOLOv5l	88.9	91.6	99.3	80.4	86.8	88.8
YOLOv7	87.6	87.7	99.1	79.3	91.6	90.0
YOLOv8x	88.1	79.0	99.1	63.1	92.4	84.9
AD-YOLO	90.2	89.3	99.4	85.9	92.3	92.5

TABLE VI
COMPARISON WITH OTHER METHODS IN THE ASS2 DATASET

Model	Precision	Recall	Aiplane	Vehicle	mAP
Faster-RCNN	17.5	11.7	10.2	0.3	5.3
SSD300	40.5	3.2	7.5	0.2	3.8
CenterNet	81.3	53.9	80.1	30.9	55.5
YOLOv3	82.4	54.8	77.7	32.6	55.2
YOLOv4	80.8	47.3	69.4	27.3	48.4
YOLOv5l	41.7	55.4	82.3	34.1	58.2
YOLOv7	84.1	52.7	74.8	36.2	55.5
YOLOv8x	80.9	49.2	77.6	38.8	58.2
AD-YOLO	78.5	68.1	87.1	51.8	69.4

the performance of AD-YOLO with Faster-RCNN, SSD300, CenterNet, YOLOv3, YOLOv4, YOLOv5l, YOLOv7, and YOLOv8x in the ASS1.

Table V shows that the proposed method achieves the best Precision, Recall, and mAP results. Compared to the YOLOv7 method, the proposed method has improved Precision, Recall, and mAP by 2.6%, 1.6%, and 2.5%, respectively. The improvement of evaluation indicators is attributed to the larger quantity of small objects on the ASS1 dataset. We use the other detectors in the ASS2 for a fair comparison to further validate the performance of AD-YOLO in detecting small objects, including Faster-RCNN, SSD300, CenterNet, YOLOv3, YOLOv4, YOLOv5l, YOLOv7, and YOLOv8x.

Table VI shows that the AD-YOLO increases by 13.9% mAP compared with the original YOLOv7. The AP of the vehicle is 51.8%, representing an improvement of 15.6% compared with the original method. Although the proposed

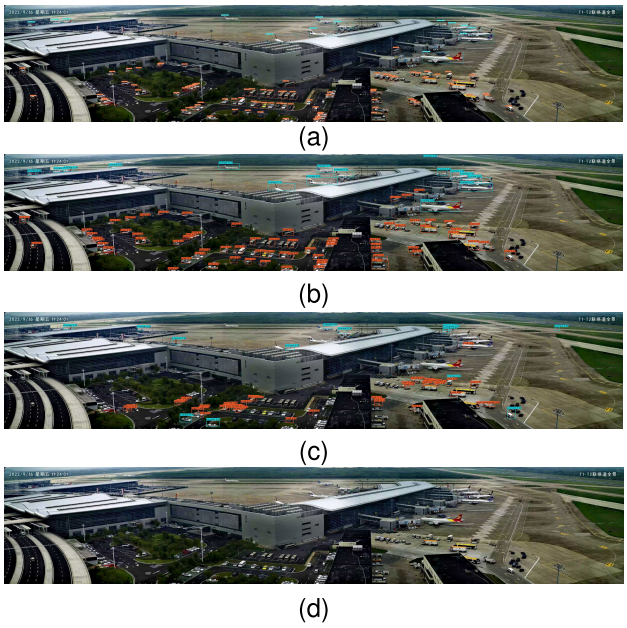


Fig. 12. Visualization of the representative detection results on the ASS2. (a) Faster-RCNN. (b) YOLOv5l. (c) YOLOv7. (d) Proposed AD-YOLO.

method still has errors in airport detection, it achieves the highest accuracy among existing detection methods. The improvement in accuracy not only significantly enhances airport security and reduces manual intervention but also increases airport operational efficiency, thereby promoting the continuous development of airport operations. It indicates that AD-YOLO is more capable of focusing on small object information and is suitable for airport scene detection tasks.

This article randomly selects one image from each ASS1 and ASS2 dataset. The representative detection methods test airport scene targets, as shown in Figs. 11 and 12. A person target is in Fig. 11, but Faster-RCNN and SSD300 methods miss detection. The detection confidence of a person is 74% in Fig. 11(d), which is the highest confidence level in detection methods. Fig. 12(d) shows the number of detected airport targets is the highest. The experiment demonstrates the excellent performance of AD-YOLO for airport scene detection.

V. CONCLUSION

In this article, we propose an airport object detection framework based on improved YOLOv7 for airport scene detection tasks. To address the problem of the existing airport datasets being single and lacking multiairport image data, we construct the MASD airport dataset with 5736 images consisting of actual and online airport videos. By analyzing airport object features, we find that the objects include many small objects. Therefore, we first design a plug-and-play ESCA to focus more on the problem of small-object omission in airport object detection by suppressing the image background. Second, the Swin Transformer module provides the backbone with more substantial feature extraction capabilities. Then, the SODL is added to boost the small-object detection capability of the network. Finally, the ECSA is inserted in different positions of the backbone and head to determine the optimal solution.

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By comparing with other methods, we demonstrate that AD-YOLO performs optimally for the general type of airport objects. The experiments show that the proposed method performs well with 71.6% of mAP in the MASD. The proposed method achieves 101.4 FPS on the NVIDIA RTX3080 GPU and 17.5 FPS on the Jetson Orin NX, maintaining a balance between accuracy and efficiency. To ensure the generalization performance of the experiments, we compare the proposed method with other detection methods and achieve the best performance in the public ASS dataset. In the future, we will collect data on different types of object data from airports to enhance the diversity of the airport scene. We will improve the detection speed of the algorithm in embedded development devices.

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